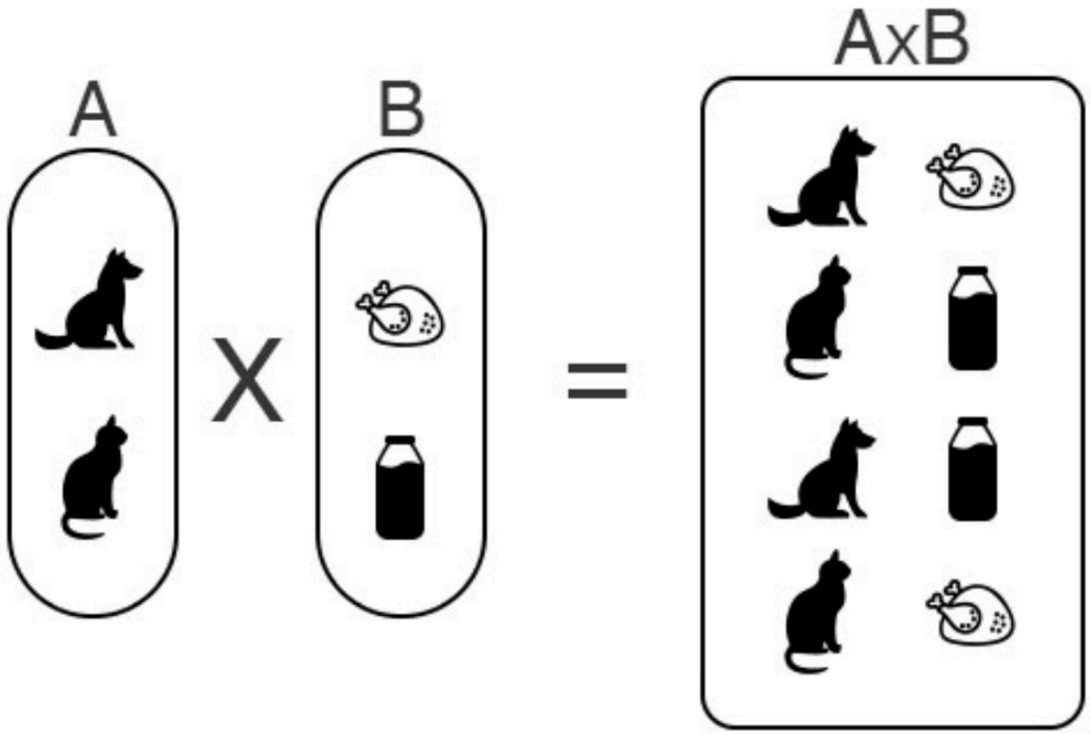
Data Literacy

Regular expressions + Sub Queries

NIFR 2023

- How to join tables (remember **ON or USING** 🤑)



Cartesian Product of Two Sets.

Recap

Working with data in MySQL

- The relational model
- Tables, Columns, Rows, MySQL Data types
- How to retrieve data
  - WHERE
- How to join tables (remember **ON or USING** 🤑)
- Aggregate functions
  - GROUP BY
  - Max/AVG/Count
- Regular expressions
- Sub-Queries

Agenda

Data Literacy

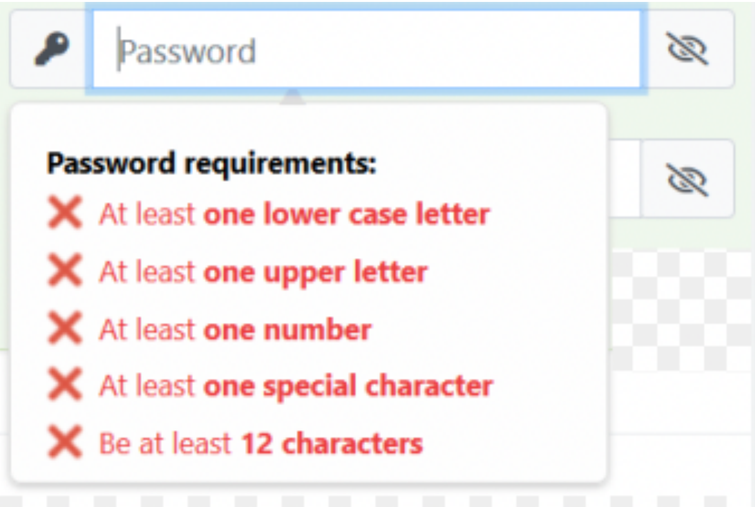
- Regular Expressions
- Regular Expressions in MySQL
- Sub Queries

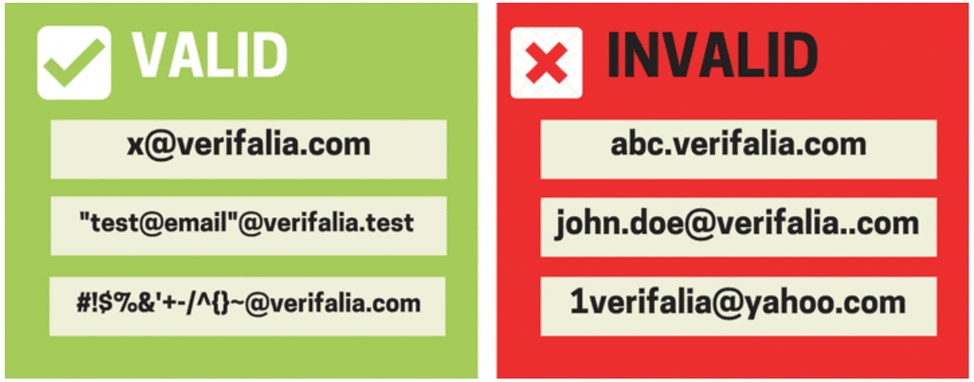
Regular Expressions RegEx

423-23-1349

Regular Expression

Validation & Search






```
const regex = /^\[(.*?)\](\(.*?\))$/;
const str = '[Google](http://www.google.com)';


const replacement = '<a href ="$2">$1</a>';

const result = str.replace(regex, replacement);

console.log(result);
```

RegEx

To match a valid social security number

/^[\d{3}-\d{2}-\d{4}]/

423-23-1349

42f-€3-1349

Regular Expression

Definition

- Text Filtering
- Pattern matching
- Generalised expression
- Universal

- Tool: https://regexr.com/

Javascript

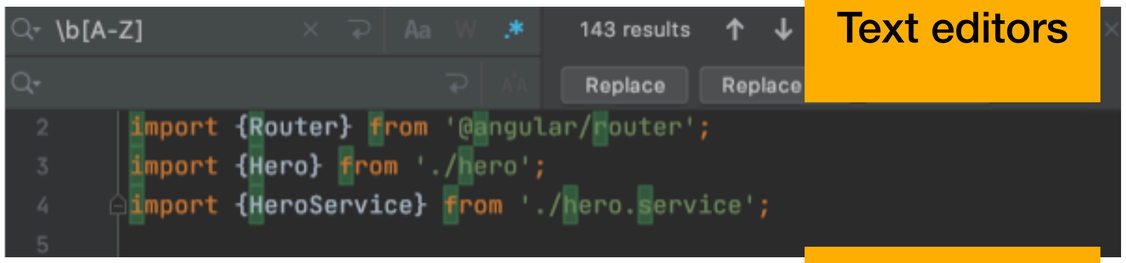
JS

```
const re = new RegExp("ab+c");
```

SQL

```
SELECT name FROM users
  WHERE name REGEXP '^[sp][aeiou]';
```

Text editors



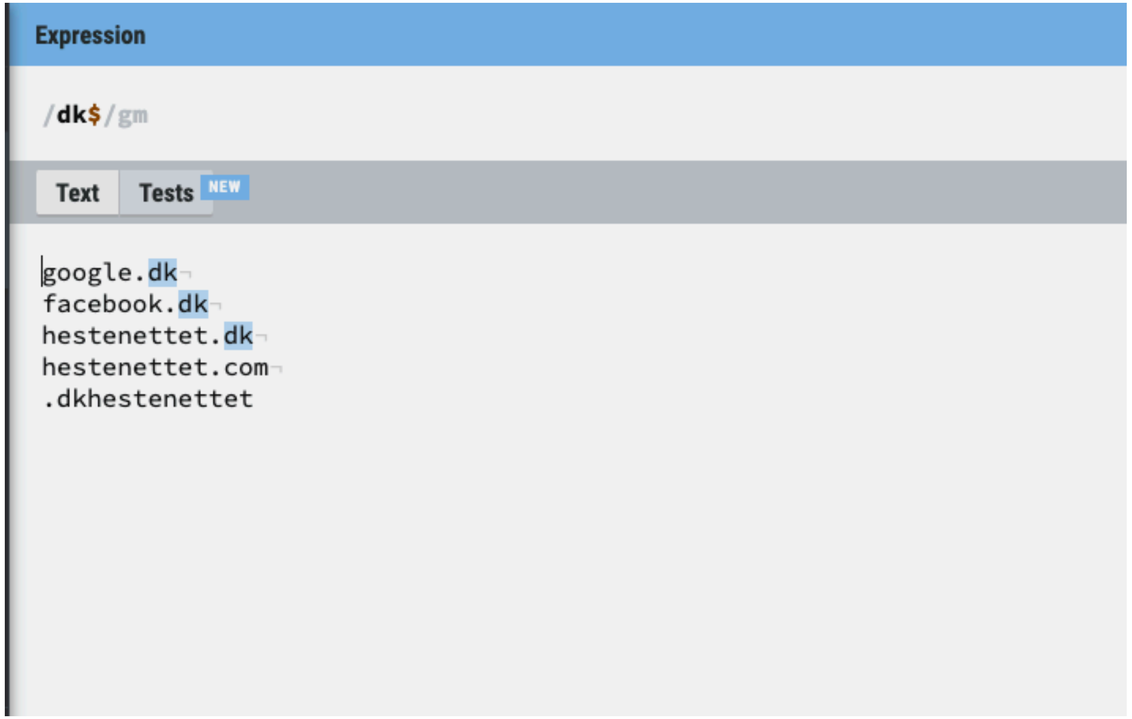
Css

```
#sections div[id^='s'] {
  color: red;
}
```

Anchors

Regular Expressions

- ^ start of string
  - ^\+45
- $ end of string
  - .dk$



https://regexr.com/7jp10

Quantifiers

Regular Expressions

- * zero or more
  - (.dk)* matches nifr@kea.dk (but also nifr@kea)

- + one or more
  - (.dk)+ matches nifr@kea.dk (but also nifr@kea.dk.dk)

- ? Zero or one

Quantifiers

Regular Expressions

- Hello{2} - matches a string with ‘hell’ and two and only two ‘o’
- Hello{2,} - matches a string with ‘hell’ and two or more ‘o’
- Hello{2,4} - matches a string with ‘hell’ and two to four ‘o’

Classes

Regular Expressions

- \d digits
  - \d{2,5} = two to five digits in a row

- \w words (strings, characters)
  - ^\w* = zero or more characters at the start of a text

- \s space (tab, space, linebreak)
  - +\d{2}\s\d{8} matches a danish phone number in the format of = +45 22334455

Flag & any

Regular Expressions

- OR operator: |
  - /(.dk|.com)$/ = danish or american domain
- . = any character

Testing

Regular Expression \d{3}-\d{2}-\d{4}

- Success cases: 113-23-4142
- Fail cases: aljnadkd
- Edge cases: 123-12-419

https://regexr.com/

Anchors

`^`	Start of string or line
`$`	End of string or line
`\b`	Word Boundary
`\B`	Not Word Boundary

Flags

`i`	Ignore Case
`g`	Global
`m`	Multiline

Group & References

`()`	Group
`\1`	Reference
`(?:)`	Non Capturing Group

Character Classes

`[abc]`	Character Set
`[^abc]`	Negated Character Set
`[a-z]`	Range
`.`	Dot
`\w`	Word
`\W`	Not Word
`\d`	Digit
`\D`	Not Digit
`\s`	Whitespace
`\S`	Not Whitespace

Lookarounds

`(?=)`	Positive Lookahead
`(?!)`	Negative Lookahead
`(?<=)`	Positive Lookbehind
`(?<!)`	Negative Lookbehind

Quantifiers And Alternation

`+`	Plus
`*`	Asterisk
`{1,3}`	Quantifier
`?`	Optional
`|`	Alternation

https://regexlearn.com/cheatsheet

Exercise 1

Regex in MySQL

```sql
SELECT *
FROM spotify
WHERE artist REGEXP '[0-9]'
```

https://www.geeksforgeeks.org/mysql-regular-expressions-regexp/

MySQL Regex

Implementation

- Classes are not written: \d, \w, \s
  - \d = [:digit:]*
  - \s = [:space:]
  - \w = [:alpha:]?
  - [:upper:] = upper case letters
- Not case sensitive

Exercise 2

What is a query?

Query: To ask a question

Response: To answer the question

Sub Queries:
Using a result to query the database

Outer Query

Subquery or Inner Query

```
SELECT lastname, firstname
FROM employees
WHERE officeCode IN (SELECT officeCode
                     FROM offices
                     WHERE country = 'USA')
```

Spotify: All above average songs in length

Exercise 3+4